Operational Semantics - Rule-Based Model Transformation

Niels Van den Broeck, Robbe Stuer

1 Introduction

We will implement the same operational semantics as in the previous assignment, but now using rule-based model transformations.

2 Modifications to meta-model

In our initial meta-model, we connected tiles to each other using directional associations: TileNorth, TileEast, TileSouth, and TileWest. This worked well, but we later realized that it is not ideal for code repeatability. For this assignment, we modified the meta-model so that only one association is used for connecting tiles: "AdjacentTile". This association has an attribute "direction", which is a string we can set to North, East, South, or West. The functionality stays the same, but we will be able to implement the moving mechanics without having to make multiple rules or adding loads of python code in constraints (unwanted).

Additionally, an attribute to creaturestate "fought" is added for the same reason as "moved". We will later check if the monster has already fought with the hero, change it to True if so, and reset it after each step, preventing endless fighting rules from occurring in a single step.

3 Status Description

For the small description about the current state we are in, we copied the code that we implemented in the previous assignment, and extended it slightly. We now also show whether the Hero or Monster moved and fought. In addition, we also show how many objective points the hero collected, making it easier to see your progress.

4 Implementation of rules

Rules are implemented and added in a certain order. Groups (a list) of rules are appended to the rule's variable to keep their order. So, at the start of a timestep, the first group of rules can be applied or 'fire'. As long as a rule is still possible to fire, all other groups of rules will not show. You can see this as a 'state' in which we are in. First, the hero movement, then collecting items, moving monsters, fighting monsters, and lastly advancing the time.

Advancing the time has already been implemented for us. It simply resets the 'moved' attribute of all creatures to False. This way, all creatures can move again in this timestep.

4.1 Moving the Hero

At the start of each step, the hero can move to all its adjacent tiles (except for obstacles, of course). Since different types of tiles have different penalties/bonuses/actions, we made a separate rule for each type:

- **move_hero_standardTile**: This is the most basic movement rule. In the left hand side, we simply check if there is a standardTile that is adjacent to the tile which the hero is standing on. If this left hand side matches, the right hand side is executed where we just move the hero to the adjacent standardTile.

- **move_Hero_trap**: The only difference with the previous rule is that we now subtract 1 live from the hero in the right hand side of the rule.
- **move_hero_door**: Again, no difference with standardTile, only that we move to a door now.
- **use_door**: This rule is a little more complicated. It does the same as before, but it also checks in the left hand side if the key that is required to open the door is present in the hero's inventory.

4.2 Collecting Items

Picking up items is even easier, we simply check if there is a item on the hero's tile, and put it in its inventory. Again, we divided this into different rules:

- **pic_up_key**: Nothing extra here, just place key in inventory.
- **pick_up_objective**: Objectives have points that must be added to the WorldState. This is done in the right hand side of the rule, where we have a constraint in which we update the 'collectedPoints' attribute by adding the new points to it.

4.3 Moving the Monster

Moving the monster is only done when it is present in the same level as the hero. This was a struggle since the left hand side did not match whenever the monster was on the same tile as the hero, or was going to move to the same tile as the hero. The problem was in this part of the rule:

Listing 1: Level matching LHS

```
standingTile:RAM_Tile
heroTile:RAM_Tile

isMonsterLevel:RAM_LevelTiles(playingLevel -> standingTile)
isHeroLevel:RAM_LevelTiles(playingLevel -> heroTile)
```

What is happening here, is that it tries to find the association of the level to a tile 2 times, but if the tile is the same, it will try to find 2 associations, but only one exists, making the left hand side not match. We solved this by adding a constraint to the level instead, checking if the hero is present in that level:

Listing 2: Improved level matching LHS

```
playingLevel:RAM_Level {
    condition = ```
        heroFound = False
        hero = get_all_instances("Hero")[0][1]
        heroT = get_target(get_outgoing(hero, "StandingTile")[0])
        for assoc in get_outgoing(this, "LevelTiles"):
            if get_target(assoc) == heroT:
                heroFound = True
        heroFound
    ```;
}

isMonsterLevel:RAM_LevelTiles(playingLevel -> standingTile)
```
```

4.3.1

In our meta-model, we specified that creatures can not stand on obstacles, also not monsters. So we need a way to make sure that the monster cannot move to an obstacle, otherwise our model becomes non-conform. This can be accomplished by adding a nac to the rule. This is a separate file that, if this matches, will not trigger the rule. If we specify in this rule that the target tile is an obstacle, It cannot match and thus this rule shall not trigger, exactly what we wanted.

4.4 Fighting

We want the monster and the hero to fight each other when they are located on the same tile. As explained before, the rules will only match if the fought attribute is set to False of the given monster. The locations of both the hero and monster are checked afterwards. If they are the same, the rule will fire. Fighting has 3 outcomes:

- **win_fight**: In the left hand side, we check who has the most lives, if the hero does, this rule can fire and the monster will lose a life.
- **lose_fight**: Otherwise, if the monster has the most lives, the hero will lose a life.
- **draw_fight**: In the other case, where they both have the same number of lives, they both lose a life.

5 Example Trace

Since one timestep now has multiple actions, traces become very large. We tried to generate a trace where most of the functionality is present. The trace is shown below. When we run the runner, the first thing we see is our status description that we described in section 3. Then the first timestep starts, we have some different moving options. We went with option a since it leads us to the key. After that, it gives us the rule to pick up the key, putting it in our inventory, which will be useful later. The monster is not in our level, so it will not be able to move, making this the end of the first timestep (we advance time, resetting moved and fought attributes). Now, we move towards the door. Again, nothing else can happen currently. In timestep 3, we use the door to go to the other level, where a monster is present. Now the monster is able to move, and we chose to move it to the tile where the hero is standing on. This goes into fighting, with only winning as the matching rule, since the hero has more lives than the monster (and thus the other rules cannot match). These were 8 steps, leaving the other 2 since they are not interesting to our trace.

```
Start simulation
Decision maker: InteractiveDecisionMaker()
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level1 at tile: tile2
  Moved: False
  Inventory:
     Empty
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: tile5
  Moved: False
  -------------
```

```
--------------
CONFORM

  a. move_hero_standardTile
      {'areAdjacent': '__AdjacentTile_67',
       'hero': 'myHero',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'isHeroState': '__CreatureState_ass_91',
       'isStandingTile': '__StandingTile_88',
       'standardTile': 'tile3',
       'standingTile': 'tile2'}
  b. move_hero_standardTile
      {'areAdjacent': '__AdjacentTile_66',
       'hero': 'myHero',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'isHeroState': '__CreatureState_ass_91',
       'isStandingTile': '__StandingTile_88',
       'standardTile': 'tile1',
       'standingTile': 'tile2'}
  c. move_hero_trap
      {'StandingTile': '__StandingTile_88',
       'areAdjacent': '__AdjacentTile_69',
       'hero': 'myHero',
       'hero.RAM_lives': 'myHero.lives',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'hero_RAM_lives': 'myHero_lives',
       'isHeroState': '__CreatureState_ass_91',
       'standingTile': 'tile2',
       'trap': 'tile4'}
Select action: a
        executed rule 'move_hero_standardTile'
          {'areAdjacent': '__AdjacentTile_67',
           'hero': 'myHero',
           'heroState': 'myHeroState',
           'heroState.RAM_moved': 'myHeroState.moved',
           'heroState_RAM_moved': 'myHeroState_moved',
           'isHeroState': '__CreatureState_ass_91',
           'newStandingTile': 'newStandingTile',
           'standardTile': 'tile3',
           'standingTile': 'tile2'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level1 at tile: tile3
  Moved: True
  Inventory:
     Empty
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
```

```
  Located in level: level2 at tile: tile5
  Moved: False
  -------------

--------------
CONFORM

  a. pick_up_key
      {'hero': 'myHero',
       'isStandingTile': 'newStandingTile',
       'key': 'key1',
       'standingTile': 'tile3',
       'tileKey': '__StandardTileItem_83'}
warning: pattern has multiple components: 2
Select action: a
        executed rule 'pick_up_key'
          {'collectKey': 'collectKey',
           'hero': 'myHero',
           'isStandingTile': 'newStandingTile',
           'key': 'key1',
           'standingTile': 'tile3'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level1 at tile: tile3
  Moved: True
  Inventory:
    key1
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: tile5
  Moved: False
  -------------

--------------
CONFORM

warning: pattern has multiple components: 2
  a. advance_time
      {'theClock': 'c', 'theClock.RAM_time': 'c.time', 'theClock_RAM_time
          ': 'c_time'}
Select action: a
        executed rule 'advance_time'
          {'theClock': 'c', 'theClock.RAM_time': 'c.time', '
              theClock_RAM_time': 'c_time'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level1 at tile: tile3
  Moved: False
  Inventory:
    key1
  Collected points: 0
```

```
  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: tile5
  Moved: False
  -------------

--------------
CONFORM

  a. move_hero_standardTile
      {'areAdjacent': '__AdjacentTile_68',
       'hero': 'myHero',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'isHeroState': '__CreatureState_ass_91',
       'isStandingTile': 'newStandingTile',
       'standardTile': 'tile2',
       'standingTile': 'tile3'}
  b. move_hero_door
      {'areAdjacent': '__AdjacentTile_71',
       'door': 'door1',
       'hero': 'myHero',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'isHeroState': '__CreatureState_ass_91',
       'isStandingTile': 'newStandingTile',
       'standingTile': 'tile3'}
Select action: b
        executed rule 'move_hero_door'
          {'areAdjacent': '__AdjacentTile_71',
           'door': 'door1',
           'hero': 'myHero',
           'heroState': 'myHeroState',
           'heroState.RAM_moved': 'myHeroState.moved',
           'heroState_RAM_moved': 'myHeroState_moved',
           'isHeroState': '__CreatureState_ass_91',
           'newStandingTile': 'newStandingTile0',
           'standingTile': 'tile3'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level1 at tile: door1
  Moved: True
  Inventory:
    key1
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: tile5
  Moved: False
  -------------

--------------
```

```
CONFORM

warning: pattern has multiple components: 2
  a. advance_time
      {'theClock': 'c', 'theClock.RAM_time': 'c.time', 'theClock_RAM_time
         ': 'c_time'}
Select action: a
        executed rule 'advance_time'
          {'theClock': 'c', 'theClock.RAM_time': 'c.time', '
             theClock_RAM_time': 'c_time'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level1 at tile: door1
  Moved: False
  Inventory:
    key1
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: tile5
  Moved: False
  -------------

--------------
CONFORM

  a. move_hero_standardTile
      {'areAdjacent': '__AdjacentTile_72',
       'hero': 'myHero',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'isHeroState': '__CreatureState_ass_91',
       'isStandingTile': 'newStandingTile0',
       'standardTile': 'tile3',
       'standingTile': 'door1'}
  b. move_hero_trap
      {'StandingTile': 'newStandingTile0',
       'areAdjacent': '__AdjacentTile_74',
       'hero': 'myHero',
       'hero.RAM_lives': 'myHero.lives',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
       'heroState_RAM_moved': 'myHeroState_moved',
       'hero_RAM_lives': 'myHero_lives',
       'isHeroState': '__CreatureState_ass_91',
       'standingTile': 'door1',
       'trap': 'tile4'}
  c. use_door
      {'areConnected': '__ConnectingDoor_81',
       'connectedDoor': 'door2',
       'doorKey': '__DoorKey_86',
       'hero': 'myHero',
       'heroKey': 'collectKey',
       'heroState': 'myHeroState',
       'heroState.RAM_moved': 'myHeroState.moved',
```

```
       'heroState_RAM_moved': 'myHeroState_moved',
       'isHeroState': '__CreatureState_ass_91',
       'key': 'key1',
       'standingDoor': 'door1',
       'standingTile': 'newStandingTile0'}
Select action: c
        executed rule 'use_door'
          {'areConnected': '__ConnectingDoor_81',
           'connectedDoor': 'door2',
           'doorKey': '__DoorKey_86',
           'hero': 'myHero',
           'heroKey': 'collectKey',
           'heroState': 'myHeroState',
           'heroState.RAM_moved': 'myHeroState.moved',
           'heroState_RAM_moved': 'myHeroState_moved',
           'isHeroState': '__CreatureState_ass_91',
           'key': 'key1',
           'newStandingTile': 'newStandingTile',
           'standingDoor': 'door1'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level2 at tile: door2
  Moved: True
  Inventory:
    key1
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: tile5
  Moved: False
  -------------

--------------
CONFORM

warning: pattern has multiple components: 2
  a. move_monster
      {'areAdjacent': '__AdjacentTile_79',
       'isMonsterLevel': '__LevelTiles_61',
       'isMonsterState': '__CreatureState_ass_92',
       'isStandingTile': '__StandingTile_89',
       'monster': 'monster',
       'monsterState': 'monsterState',
       'monsterState.RAM_moved': 'monsterState.moved',
       'monsterState_RAM_moved': 'monsterState_moved',
       'playingLevel': 'level2',
       'standingTile': 'tile5',
       'targetTile': 'tile7'}
  b. move_monster
      {'areAdjacent': '__AdjacentTile_76',
       'isMonsterLevel': '__LevelTiles_61',
       'isMonsterState': '__CreatureState_ass_92',
       'isStandingTile': '__StandingTile_89',
       'monster': 'monster',
       'monsterState': 'monsterState',
       'monsterState.RAM_moved': 'monsterState.moved',
```

```
       'monsterState_RAM_moved': 'monsterState_moved',
       'playingLevel': 'level2',
       'standingTile': 'tile5',
       'targetTile': 'door2'}
Select action: b
        executed rule 'move_monster'
          {'areAdjacent': '__AdjacentTile_76',
           'isMonsterLevel': '__LevelTiles_61',
           'isMonsterState': '__CreatureState_ass_92',
           'monster': 'monster',
           'monsterState': 'monsterState',
           'monsterState.RAM_moved': 'monsterState.moved',
           'monsterState_RAM_moved': 'monsterState_moved',
           'newStandingTile': 'newStandingTile0',
           'playingLevel': 'level2',
           'standingTile': 'tile5',
           'targetTile': 'door2'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: False
  Located in level: level2 at tile: door2
  Moved: True
  Inventory:
    key1
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 5
  Fought: False
  Located in level: level2 at tile: door2
  Moved: True
  -------------

--------------
CONFORM

warning: pattern has multiple components: 2
  a. win_fight
      {'hero': 'myHero',
       'hero.RAM_lives': 'myHero.lives',
       'heroState': 'myHeroState',
       'heroState.RAM_fought': 'myHeroState.fought',
       'heroState_RAM_fought': 'myHeroState_fought',
       'heroTile': 'newStandingTile',
       'hero_RAM_lives': 'myHero_lives',
       'isHeroState': '__CreatureState_ass_91',
       'isMonsterState': '__CreatureState_ass_92',
       'monster': 'monster',
       'monster.RAM_lives': 'monster.lives',
       'monsterState': 'monsterState',
       'monsterState.RAM_fought': 'monsterState.fought',
       'monsterState_RAM_fought': 'monsterState_fought',
       'monsterTile': 'newStandingTile0',
       'monster_RAM_lives': 'monster_lives',
       'tile': 'door2'}
Select action: a
        executed rule 'win_fight'
          {'hero': 'myHero',
           'hero.RAM_lives': 'myHero.lives',
```

```
            'heroState': 'myHeroState',
            'heroState.RAM_fought': 'myHeroState.fought',
            'heroState_RAM_fought': 'myHeroState_fought',
            'heroTile': 'newStandingTile',
            'hero_RAM_lives': 'myHero_lives',
            'isHeroState': '__CreatureState_ass_91',
            'isMonsterState': '__CreatureState_ass_92',
            'monster': 'monster',
            'monster.RAM_lives': 'monster.lives',
            'monsterState': 'monsterState',
            'monsterState.RAM_fought': 'monsterState.fought',
            'monsterState_RAM_fought': 'monsterState_fought',
            'monsterTile': 'newStandingTile0',
            'monster_RAM_lives': 'monster_lives',
            'tile': 'door2'}
--------------
  ---HERO---
  Name: myHero
  Lives: 10
  Fought: True
  Located in level: level2 at tile: door2
  Moved: True
  Inventory:
    key1
  Collected points: 0

  ---MONSTERS---
  Name: monster
  Lives: 4
  Fought: True
  Located in level: level2 at tile: door2
  Moved: True
  -------------

--------------
CONFORM
```